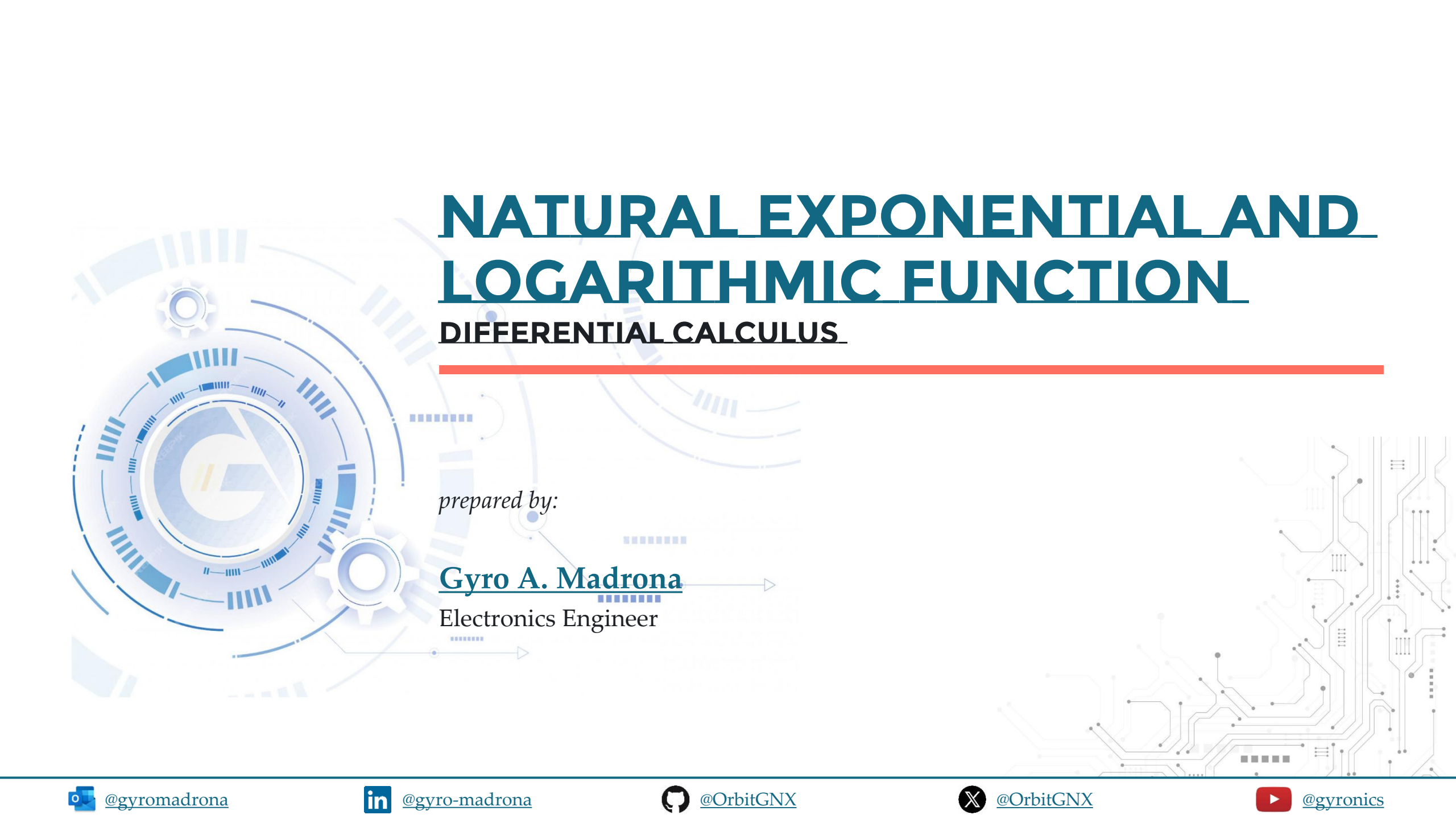




NATURAL EXPONENTIAL AND LOGARITHMIC FUNCTION

DIFFERENTIAL CALCULUS

prepared by:

Gyro A. Madrona

Electronics Engineer

TOPIC OUTLINE

Derivative of Natural Exponential Function

Derivative of Natural Logarithmic Function



DERIVATIVE OF NATURAL EXPONENTIAL FUNCTION



NATURAL EXPONENTIAL FUNCTION

Derivative of e^x

$$\frac{d}{dx}(e^x) = e^x$$

The natural exponential function is a function of the form

$$y = e^x$$

where

$e \approx 2.71828$ is the Euler's number

Chain Rule

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$



EXERCISE

Differentiate $y = e^{3x}$.

Solution

$$\frac{dy}{dx} = e^u \frac{du}{dx}$$

$$\frac{dy}{dx} = e^{3x} \frac{d}{dx}(3x)$$

$$\frac{dy}{dx} = 3e^{3x}$$

ans



EXERCISE

Differentiate $y = e^{x^2}$.

Solution

$$\frac{dy}{dx} = e^u \frac{du}{dx}$$

$$\frac{dy}{dx} = e^{x^2} \frac{d}{dx}(x^2)$$

$$\frac{dy}{dx} = 2xe^{x^2}$$

ans



EXERCISE

Differentiate $f(x) = (x^3 + 2x)e^x$.

Solution

$$f'(x) = (x^3 + 2x) \frac{d}{dx} e^x + e^x \frac{d}{dx} (x^3 + 2x)$$

$$f'(x) = (x^3 + 2x) e^x + e^x (3x^2 + 2)$$

$$f'(x) = e^x (x^3 + 3x^2 + 2x + 2)$$



EXERCISE

Differentiate $y = \frac{e^x}{1-e^x}$.

Solution

$$\frac{dy}{dx} = \frac{(1-e^x) \frac{d}{dx}(e^x) - e^x \frac{d}{dx}(1-e^x)}{(1-e^x)^2}$$

$$\frac{dy}{dx} = \frac{(1-e^x)e^x - e^x(-e^x)}{(1-e^x)^2}$$

$$\frac{dy}{dx} = \frac{e^x - e^{2x} + e^{2x}}{(1-e^x)^2}$$

$$\boxed{\frac{dy}{dx} = \frac{e^x}{(1-e^x)^2}}$$

ans



EXERCISE

Differentiate $y = e^{\tan x}$.

Solution

$$\frac{dy}{dx} = e^u \frac{du}{dx}$$

$$\frac{dy}{dx} = e^{\tan x} \frac{d}{dx}(\tan x)$$

$$\frac{dy}{dx} = e^{\tan x} (\sec^2 x)$$

$$\boxed{\frac{dy}{dx} = \sec^2 x e^{\tan x}}$$

ans



EXERCISE

Find y' if $y = e^{-4x} \sin 5x$.

Solution

$$y' = e^{-4x} \frac{d}{dx}(\sin 5x) + \sin 5x \frac{d}{dx}(e^{-4x})$$

$$y' = e^{-4x} (5 \cos 5x) + \sin 5x (-4e^{-4x})$$

$$y' = e^{-4x} (5 \cos 5x - 4 \sin 5x)$$

ans



DERIVATIVE OF NATURAL LOGARITHMIC FUNCTIONS



NATURAL LOGARITHMIC FUNCTION

Derivative of $\ln x$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

The natural logarithmic function is a function of the form

$$y = \ln(x)$$

where

$\ln x$ is the natural logarithm ($\log_e x = \ln x$)

Chain Rule

$$\frac{d}{dx}(\ln u) = \frac{1}{u} \frac{du}{dx}$$



EXERCISE

Differentiate $y = \ln(3x)$.ⁿ

Solution

$$\frac{dy}{dx} = \frac{1}{u} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{3x} \cdot \frac{d}{dx}(3x)$$

$$\frac{dy}{dx} = \frac{1}{\cancel{3x}} (\cancel{3})$$

$$\boxed{\frac{dy}{dx} = \frac{1}{x}}$$

ans



EXERCISE

Find y' if $y = \ln(x^2)$.

Solution

$$y' = \frac{1}{u} \cdot \frac{du}{dx}$$

$$y' = \frac{1}{x^2} \cdot \frac{d}{dx}(x^2)$$

$$y' = \frac{1}{x^2} (2x)$$

$$y' = \frac{2}{x}$$

ans

EXERCISE

Differentiate $y = \ln(x + 1)$. ⁿ

Solution

$$\frac{dy}{dx} = \frac{1}{u} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{(x+1)} \cdot \frac{d}{dx}(x+1)$$

$$\boxed{\frac{dy}{dx} = \frac{1}{(x+1)}}$$

ans



EXERCISE

Find y' if $y = \ln(\sqrt{x})$. h

Solution

$$y' = \frac{1}{u} \cdot \frac{du}{dx}$$

$$y' = \frac{1}{\sqrt{x}} \cdot \frac{d}{dx}(x^{\frac{1}{2}})$$

$$y' = \frac{1}{x^{\frac{1}{2}}} \cdot \left(\frac{1}{2} x^{-\frac{1}{2}} \right) \rightarrow -\frac{1}{2} - \frac{1}{2} = -1$$

$$y' = \frac{1}{2x}$$

ans



EXERCISE

Differentiate $y = \ln(\sin x)$. u

Solution

$$\frac{dy}{dx} = \frac{1}{u} \cdot \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot \frac{d}{dx}(\sin x)$$

$$\frac{dy}{dx} = \frac{1}{\sin x} \cdot (\cos x)$$

$$\boxed{\frac{dy}{dx} = \cot x}$$

ans



EXERCISE

Differentiate $f(x) = \sqrt{\ln x}$. u

$$f'(x) = \frac{du}{du} \cdot \frac{du}{dx}$$

$$f'(x) = \frac{d}{du}(\sqrt{u}) \cdot \frac{d}{dx}(\ln x)$$

$$f'(x) = \frac{1}{2} u^{-\frac{1}{2}} \cdot \left(\frac{1}{x}\right)$$

$$f'(x) = \frac{1}{2\sqrt{u}} \cdot \frac{1}{x}$$

Solution

$$f'(x) = \frac{1}{2x\sqrt{\ln x}}$$

ans



EXERCISE

Differentiate $f(x) = \sqrt{x} \ln x$.

Solution

$$f'(x) = \sqrt{x} \frac{d}{dx}(\ln x) + \ln x \frac{d}{dx}(\sqrt{x})$$

$$f'(x) = \sqrt{x} \left(\frac{1}{x} \right) + \ln x \left(\frac{1}{2\sqrt{x}} \right)$$

$$f'(x) = \frac{1}{\sqrt{x}} \cdot \frac{2}{2} + \frac{\ln x}{2\sqrt{x}}$$

$$f'(x) = \frac{2 + \ln x}{2\sqrt{x}}$$

ans



EXERCISE

Differentiate $f(x) = x^2 \ln x$.

Solution

$$f'(x) = x^2 \frac{d}{dx}(\ln x) + \ln x \frac{d}{dx}(x^2)$$

$$f'(x) = x^2 \left(\frac{1}{x} \right) + \ln x (2x)$$

$$f'(x) = x + 2x \ln x$$

ans



LABORATORY

